FPGA/GPU inspection benchmarks

Final campaign: 21 October 2026 · Report: 21 October 2026

This report compares five implemented inspection paths on the same host: Buffering, Streaming, Resource-optimized streaming, GPU_HOST, and GPU_P2P. The GPU paths use the original FP32 model; FPGA paths use the quantized W8A8 model. These are system comparisons, not equal-precision kernel comparisons.

Main result. Resource-optimized streaming reduces routed wrapper BRAM by 83.95% and DSPs by 51.20% versus Streaming. Its host in-memory latency at 4 MiB is 1.119 ms, 1.43% above Streaming. GPU_HOST / checker latency ratios are 1.84× at 4 MiB and 1.00× at the largest frame; ratios above one favor the checker.

Boundary workload. At 4,194,240 bytes, Resource-optimized streaming takes 16.824 ms and GPU_HOST takes 2.155 ms. Adjacent sizes exercise a different FPGA preprocessing path. This workload is retained in the full report and excluded from the focused report.

Variant	4 MiB host in-memory median / p95 (ms)	4 MiB file-inclusive median / p95 (ms)
Buffering	1.431 / 1.436	2.474 / 2.651
Streaming	1.104 / 1.105	2.154 / 2.287
Resource-optimized streaming	1.119 / 1.121	2.158 / 2.357
GPU_HOST	2.054 / 2.153	3.234 / 3.467
GPU_P2P	2.894 / 3.121	4.075 / 4.307

P2P outcome. GPU_P2P is slower than GPU_HOST in this setup. At the largest frame, its host in-memory median is 35.571 ms versus 10.537 ms for GPU_HOST. The measured P2P forwarding interval is 32.675 ms. Autonomous polling passed board admission; that establishes correctness, not a latency advantage.

Evidence: 30,000 primary measured calls, 6,000 closing-baseline calls, and 600 separate GPU profiling calls. Maximum accepted baseline drift: 3.84%. Runtime revision: preprocessing-v2 only; exploratory later revisions are excluded.

Reading guide: pages 3–7 explain GPU timing; pages 8–10 compare FPGA variants; pages 11–12 compare Resource-optimized streaming with GPU_HOST. Resources, validation, and complete tables follow.

Measurement protocol and interpretation

Report label	Iteration / implementation
Buffering	OPT_0_NONSTREAMING; buffers all 65,536 pixels before CNN consumption
Streaming	OPT_1_STREAMING; original CNN with streaming preprocessing
Resource-optimized streaming	OPT_6_BRAM_FIFO1024_DSP_B2_RF72_RF288; native FIFO profiling capacity 1024; conv3/4/5/6 reuse 72/72/288/288
GPU_HOST	OPT_7_GPU_HOST; host sends input to GPU and FPGA; host relays GPU result
GPU_P2P	OPT_8_GPU_P2P; FPGA forwards input to GPU and autonomously polls result mailbox

Machine: hacc-gpu-u55c-01; U55C device2, PCIe 0000:81:00.0, serial XFL13BA4HVZYA; MI210 GPU0, PCIe 0000:03:00.0. Host thread pinned to CPU64/NUMA1; GPU is on NUMA0. FPGA link: PCIe Gen3 ×16. All comparisons use the same devices and 128 MiB frame allocation.

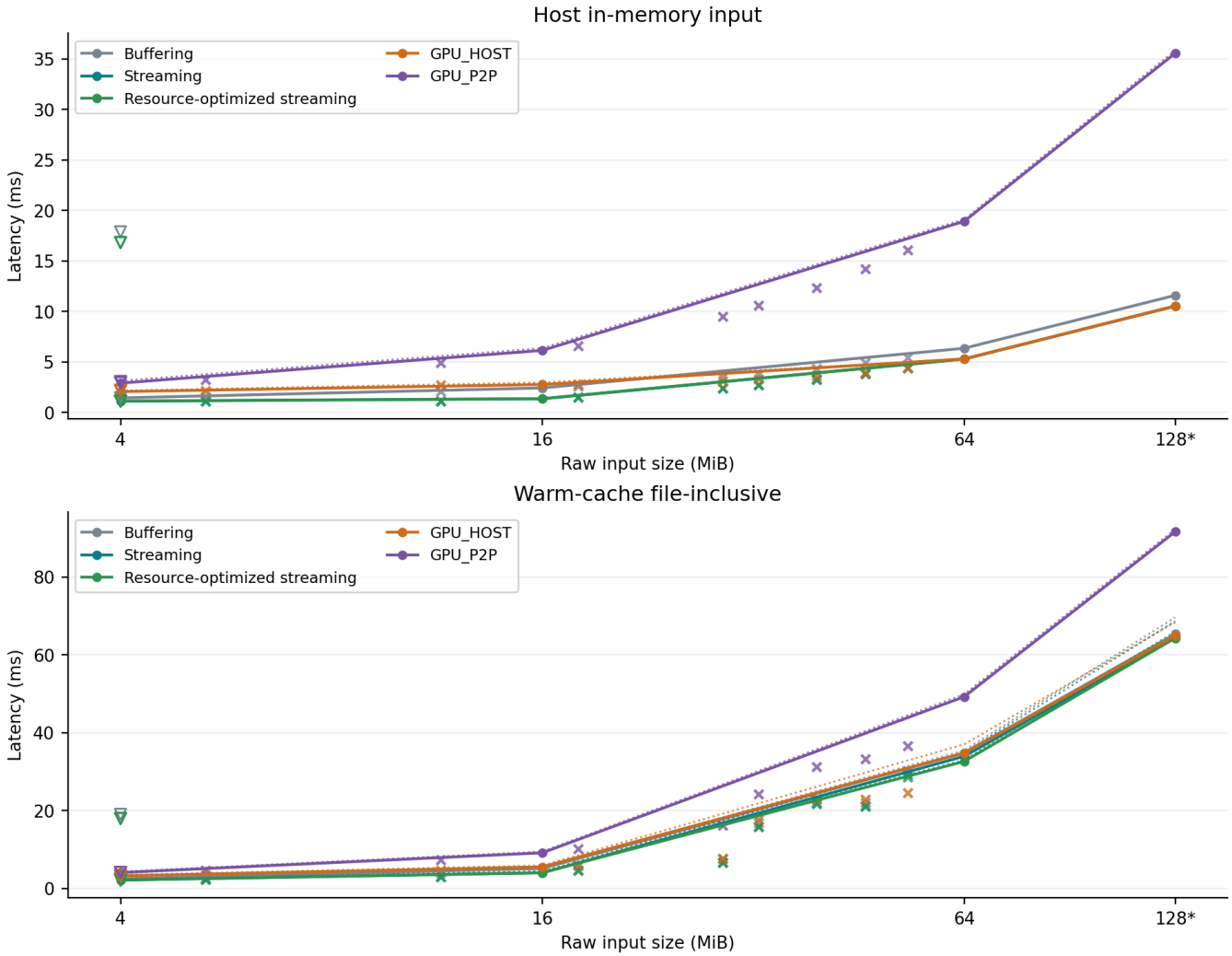
Workloads: eight real bitstreams, four synthetic sizes, and three preprocessing boundary inputs. Each variant has 20 warmups and 200 measured trials per workload per scope. Each image passes 520 board-admission calls and a pilot. The sequence is Streaming → Buffering → Resource-optimized streaming → GPU_HOST → GPU_P2P → Streaming repeat.

Host in-memory: prepared input invocation to host-observed verdict; excludes file reading, preparation, initial allocation, programming, and diagnostic CSR readback. **File-inclusive:** file open/read through preparation/copy to host-observed verdict, with a warm filesystem cache. Every call transfers the entire raw input.

Statistics: solid lines show medians; dotted lines show p95 unless stated otherwise. Synthetic points are connected; real inputs use crosses and boundary cases use hollow triangles. The final synthetic point is 134,217,664 bytes (128 MiB minus a 64-byte header). Breakdown stacks show means of consecutive intervals, not sums of medians.

Interpretation limits: streaming intervals overlap. FPGA CNN first-input-to-score includes input stalls. GPU diagnostic traces add profiling and explicit synchronization; they are not interchangeable with unprofiled E2E. Samples quantify within-run variation; no independent-day confidence interval or deployment-level accuracy equivalence is claimed.

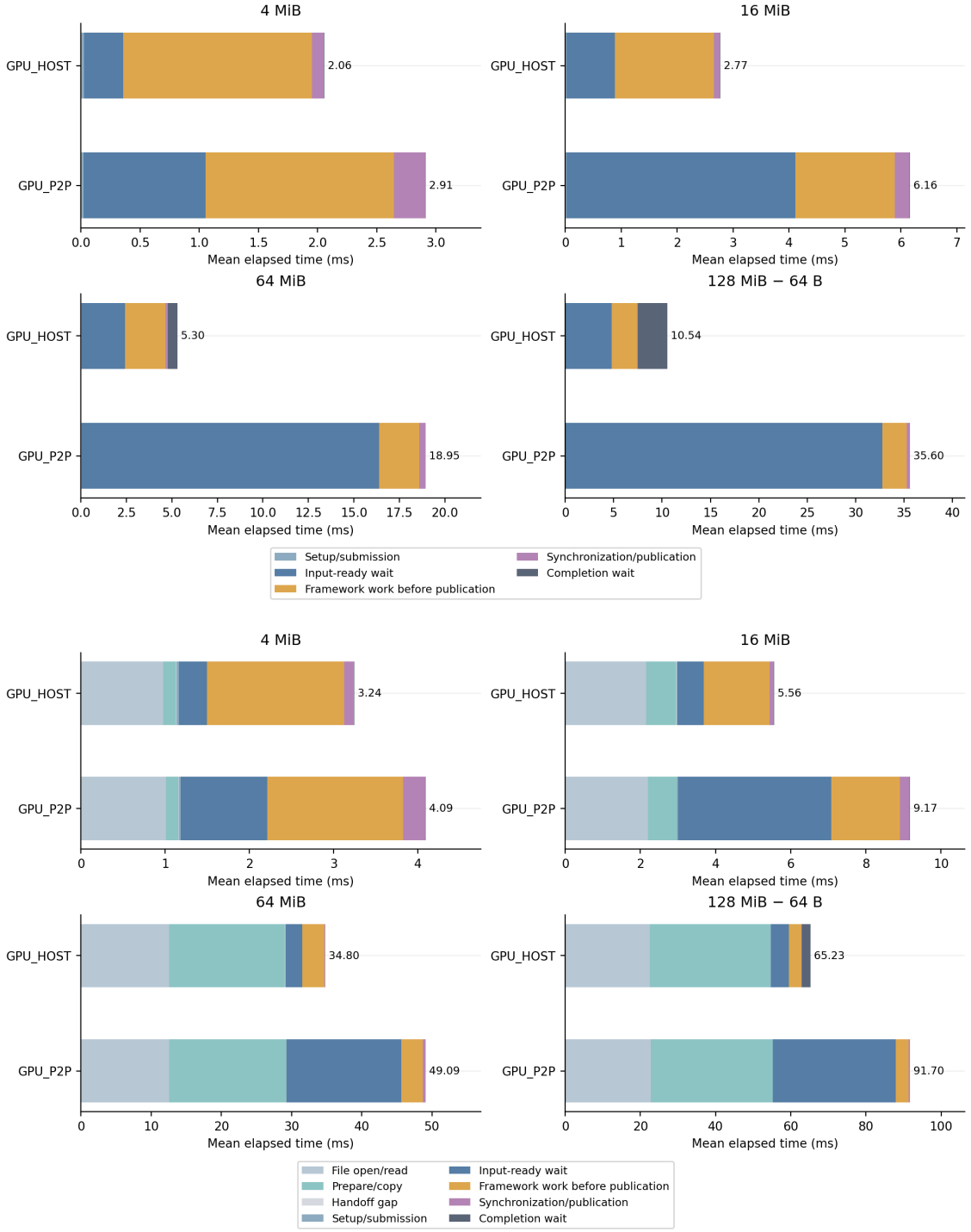
Latency across the complete workload range



The curves compare the five measured implementations for matching inputs and timing scopes. Streaming overlaps input arrival and CNN work; Buffering exposes compute after preprocessing. The corrected GPU runtime can change which implementation leads at a given size. Read individual comparisons rather than extrapolating a universal winner.

Real inputs are separate workloads, not interpolation checkpoints for the synthetic curve. The three nearly coincident boundary sizes are expanded on page 14; their p95 values are in the full tables. Dotted p95 curves apply to synthetic inputs and describe tail latency, not confidence bands.

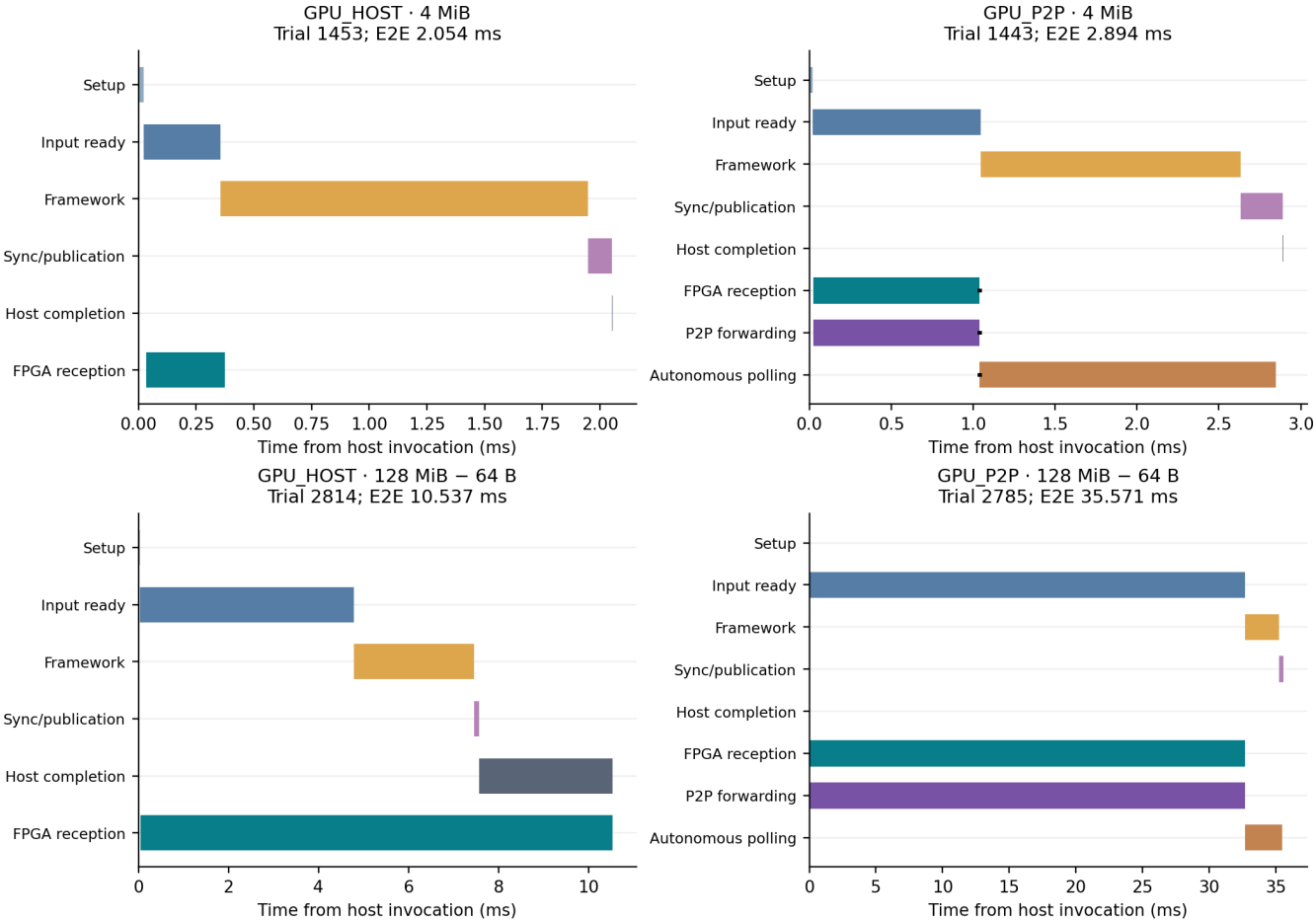
GPU paths: additive elapsed-time breakdowns



Top: host in-memory; bottom: warm-cache file-inclusive. Each stack sums exactly to its mean total. Setup includes mailbox initialization and control preparation. Input-ready wait includes transfer submission/waiting. Framework work ends when result publication begins. Synchronization/publication can include outstanding model work, score conversion, and transport; it is not a pure result-DMA measurement.

Tiny setup, handoff, and completion segments are retained even when visually narrow. Phase means and source references are exported in data/phase_means.csv.

GPU execution timelines and overlap

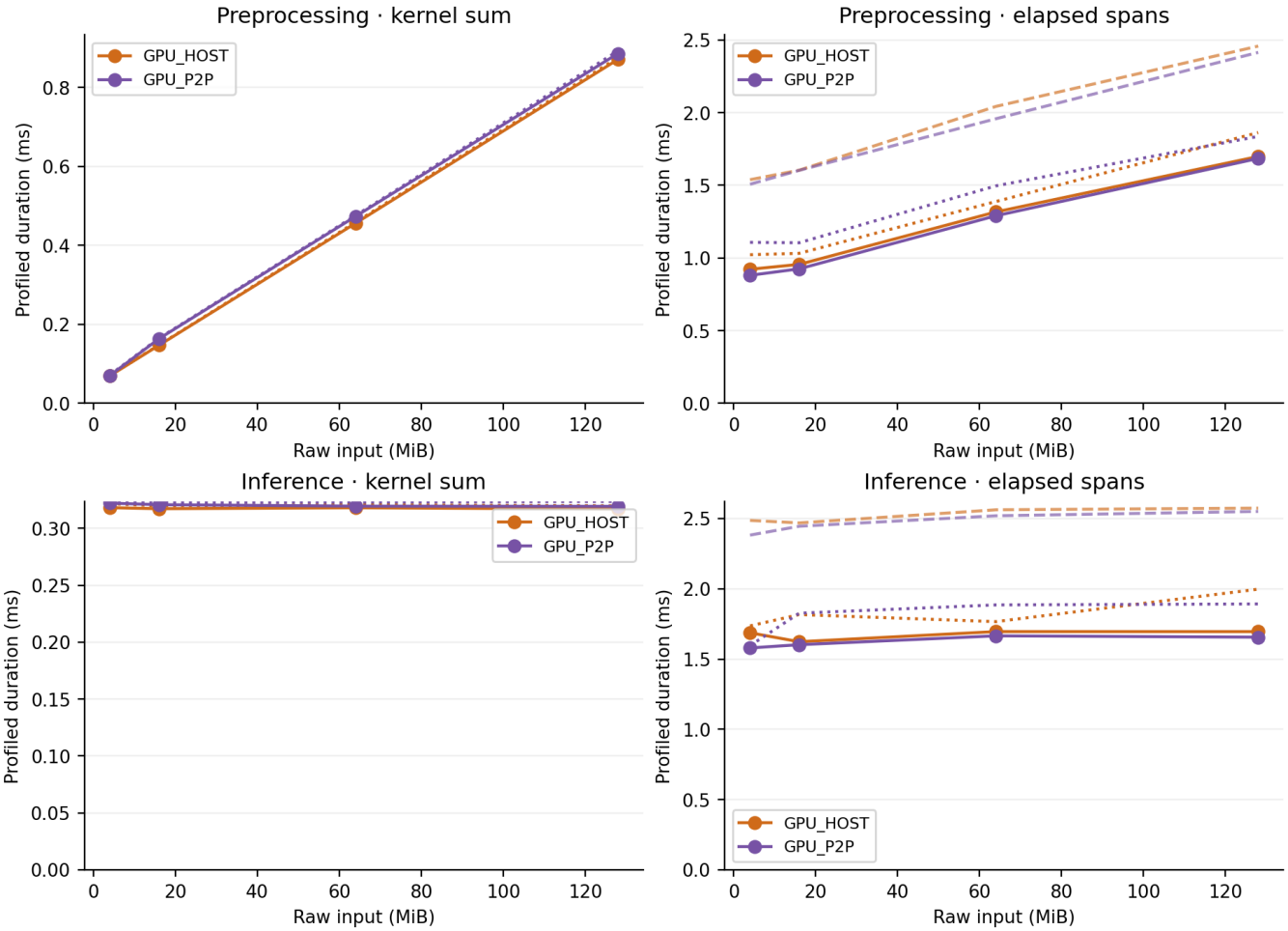


Each panel uses one actual host in-memory trial nearest its workload's median E2E. Host phases and calibrated FPGA intervals share the host-invocation origin. Black endpoint marks show feasible clock-mapping bounds; these are usually smaller than the plotted line width. Exact trial IDs and intervals are exported.

GPU_HOST submits its host-to-GPU copy and FPGA input transfer from the same host call; input-ready means the GPU copy completed, not necessarily that the FPGA accepted its final byte. In P2P, forwarding and reception overlap. FPGA polling overlaps GPU/framework work and publication; adding the polling window to these phases would double-count elapsed time.

The host-to-GPU HIP copy duration is reported separately on page 7: its host-aligned start was not recorded, so no artificial start time is drawn here.

GPU preprocessing and inference diagnostics

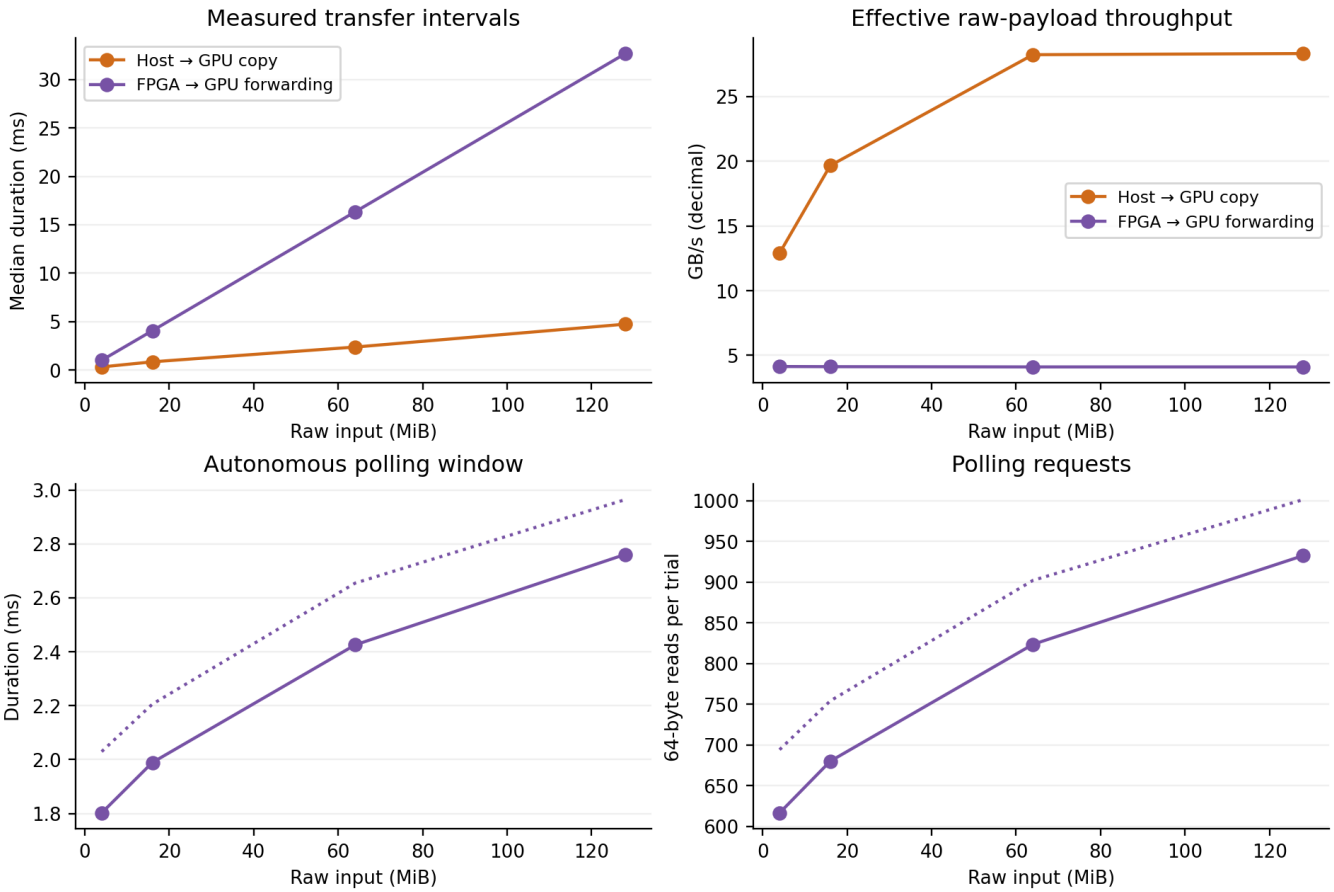


Left: sum of attributed GPU kernel durations. Right: first-to-last-kernel span, with synchronized host-phase median dashed. Twenty diagnostic calls per workload/path are separate from primary measurements. Kernel counts come from the new trace tables; preprocessing uses FP32 gather on GPU.

Kernel sums exclude copies and gaps; the device span can contain both. This revision removes the confirmed CPU uint8-gather fallback by casting bytes to FP32 before sampling. Trace checks reject input/index copies to CPU GatherV2. Diagnostic output materialization remains outside the main measurement protocol.

These profiles cannot partition an unprofiled E2E bar into exact preprocessing and inference fractions. Profiling phases were explicitly synchronized; first-use effects and profiling overhead can affect their p95 values. No profiled call is included in headline E2E statistics.

P2P transport and autonomous polling

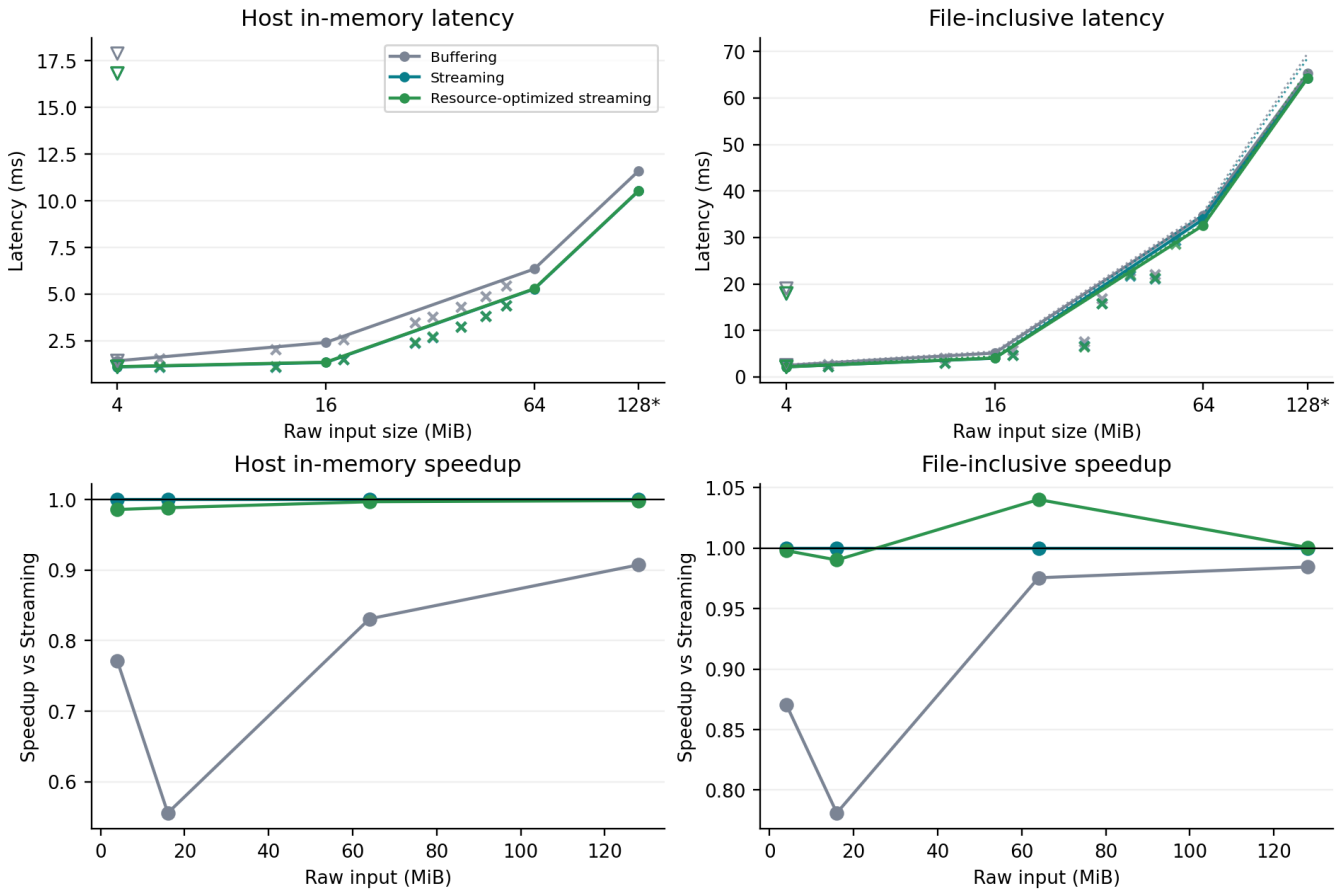


At the largest input, GPU_P2P forwards data in 32.675 ms; the GPU_HOST HIP copy interval is 4.741 ms. Effective bandwidth divides raw payload bytes by the median interval. These are different transport boundaries: P2P includes the FPGA forwarding path; the host-copy interval is measured by HIP events.

The autonomous polling window starts with FPGA polling and ends when the result payload arrives. It includes waiting for the GPU verdict; it is not isolated polling overhead. Each request reads 64 bytes. Request counts and durations describe observed traffic, not GPU compute utilization.

Correctness evidence includes 26 deliberately delayed publications during 520-call P2P admission, repeated autonomous polls, and a quiescent polling engine at completion. No poll-engine error was accepted. Cross-root/NUMA topology is documented, but these measurements do not isolate which hardware mechanism causes the transfer penalty.

Buffering, Streaming, and resource optimization

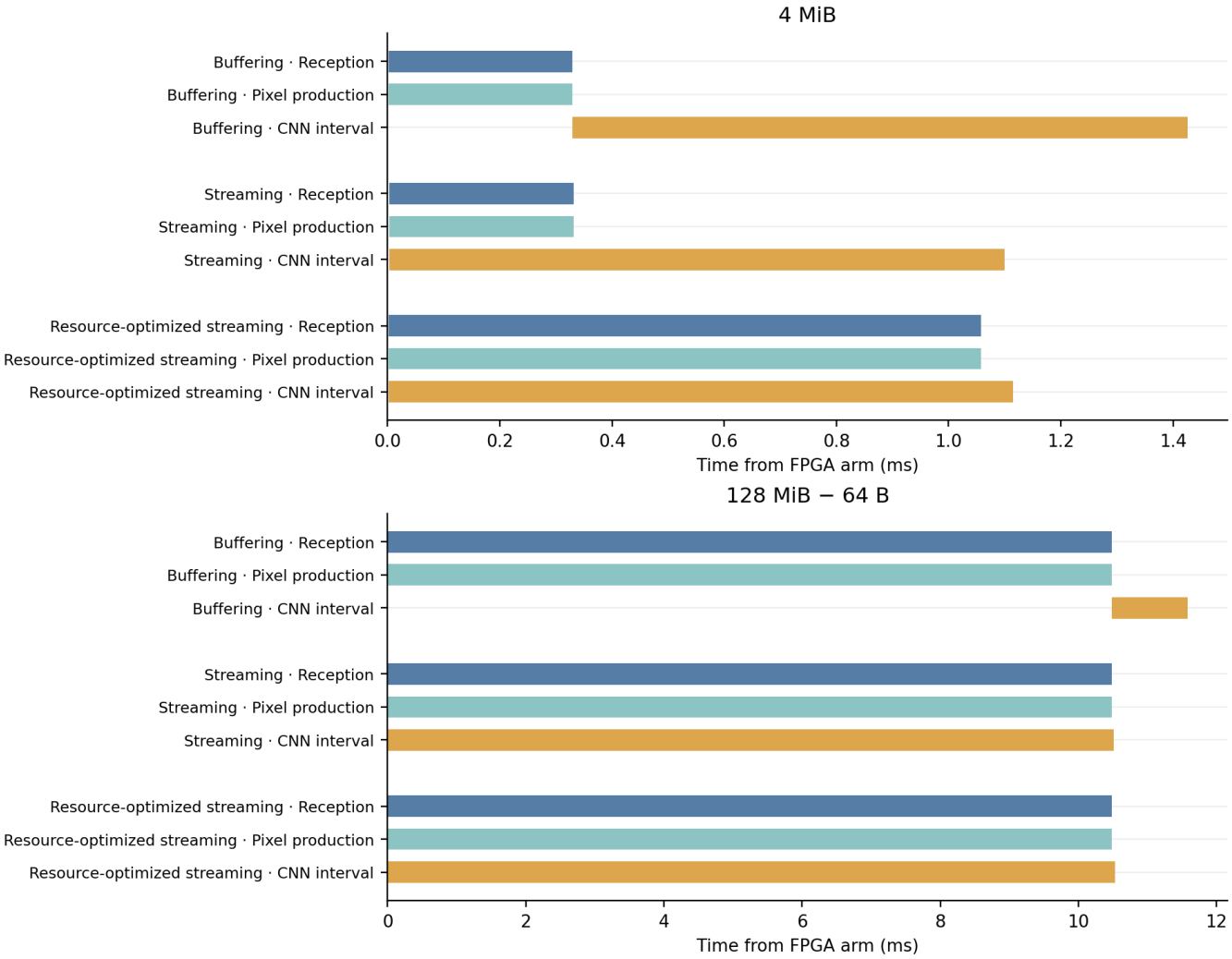


Speedup is Streaming latency divided by the compared variant's latency. Above 1 means faster than Streaming; below 1 means slower. Median speedups are ratios of medians from independent runs, not paired per-trial ratios.

At 4 MiB, Streaming is 1.30× faster than Buffering for host in-memory input. At the largest frame it is 1.10× faster. Resource-optimized streaming adds 15.8 µs at 4 MiB and 15.3 µs at the largest frame.

Small file-inclusive differences should not be interpreted as hardware wins: file read/copy variability can be larger than the FPGA invocation difference. The baseline repeat and rechecks constrain drift but do not remove all host variability.

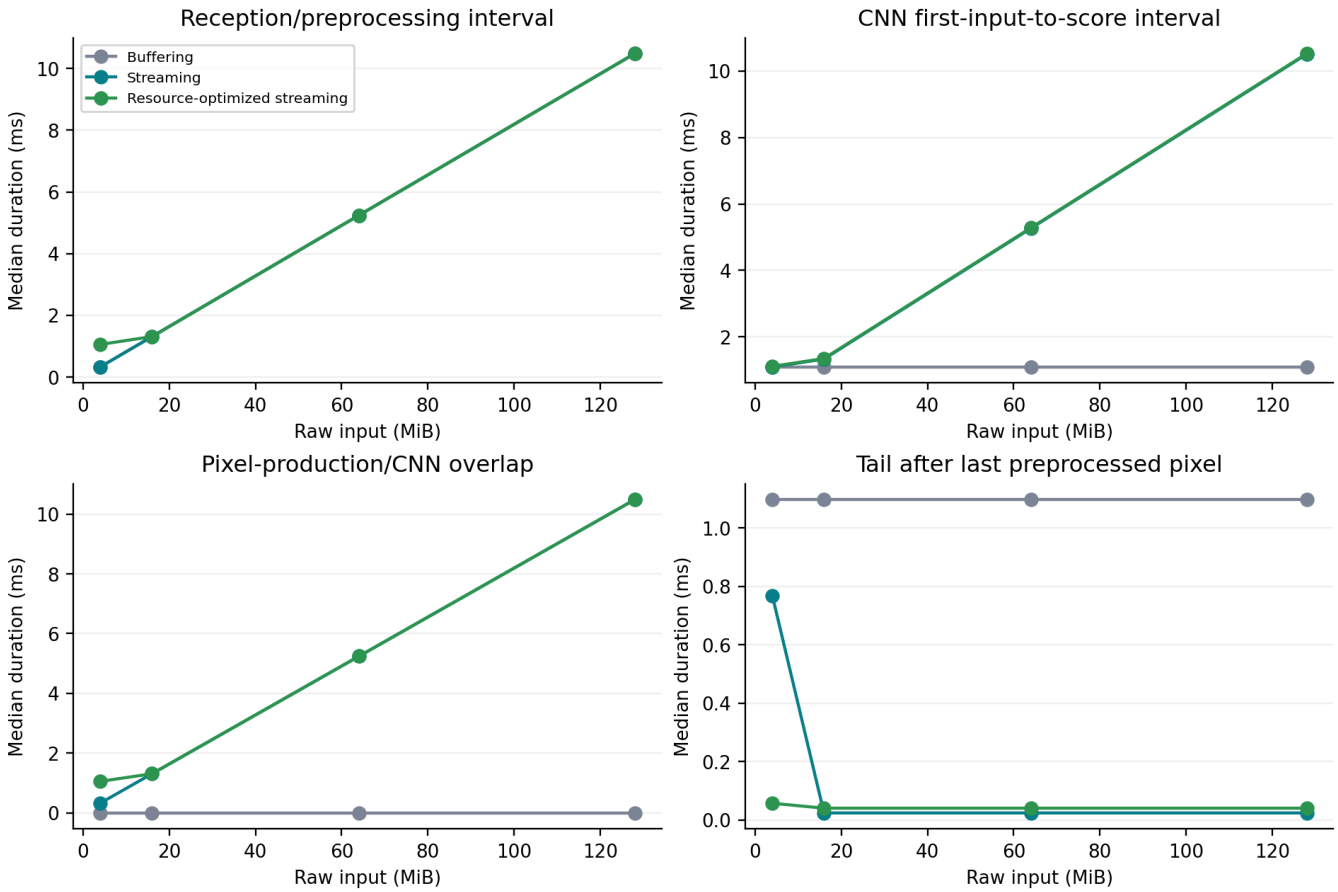
How streaming hides the CNN interval



Timelines use actual host in-memory trials nearest the median total. All lanes on this page use the FPGA clock, relative to arm; no host/FPGA alignment is needed. Pixel production starts at the first output pixel. Reception/preprocessing can begin earlier.

Buffering waits for the complete input and pixel frame before CNN consumption. Streaming begins consumption early, hiding much of the CNN interval behind input arrival. Resource-optimized streaming preserves that overlap. These are observed timelines: the CNN lane includes periods waiting for incoming pixels.

FPGA pipeline intervals across input sizes

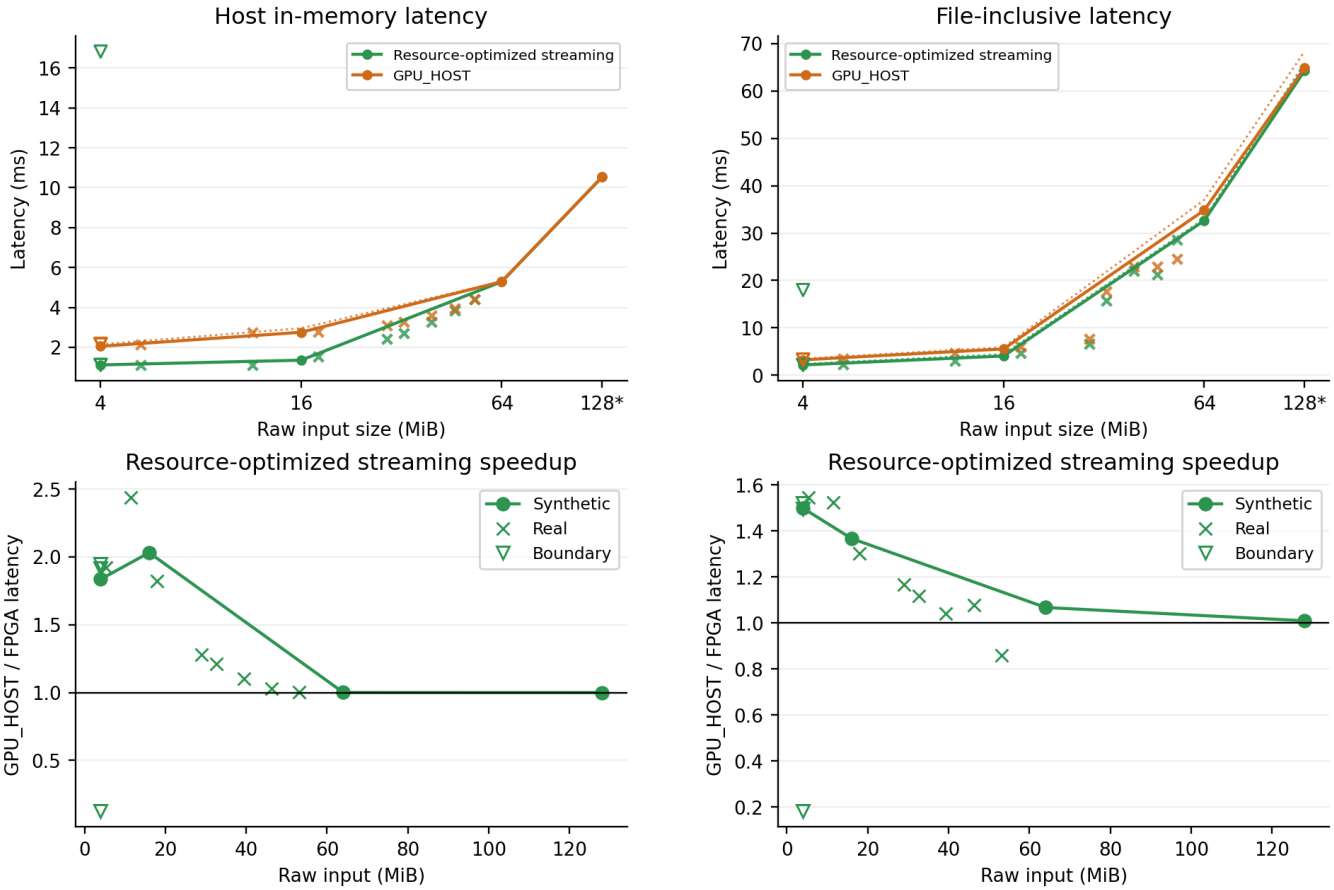


The reception/preprocessing interval runs from first raw input acceptance to last produced pixel. The CNN interval runs from first consumed pixel to score. The overlap metric covers the intersection of pixel production and CNN execution. The tail runs from last produced pixel to FPGA verdict; it is not standalone CNN compute.

At large inputs, a long streaming CNN interval mainly reflects its duration along the arriving stream. The exposed tail is much shorter than the full interval. Buffering retains an approximately fixed post-preprocessing CNN cost. FIFO pressure and input stalls can change observed phase lengths without causing failure.

These four quantities have different boundaries and overlap. They are plotted as separate curves rather than summed into a misleading total.

Resource-optimized streaming versus GPU_HOST



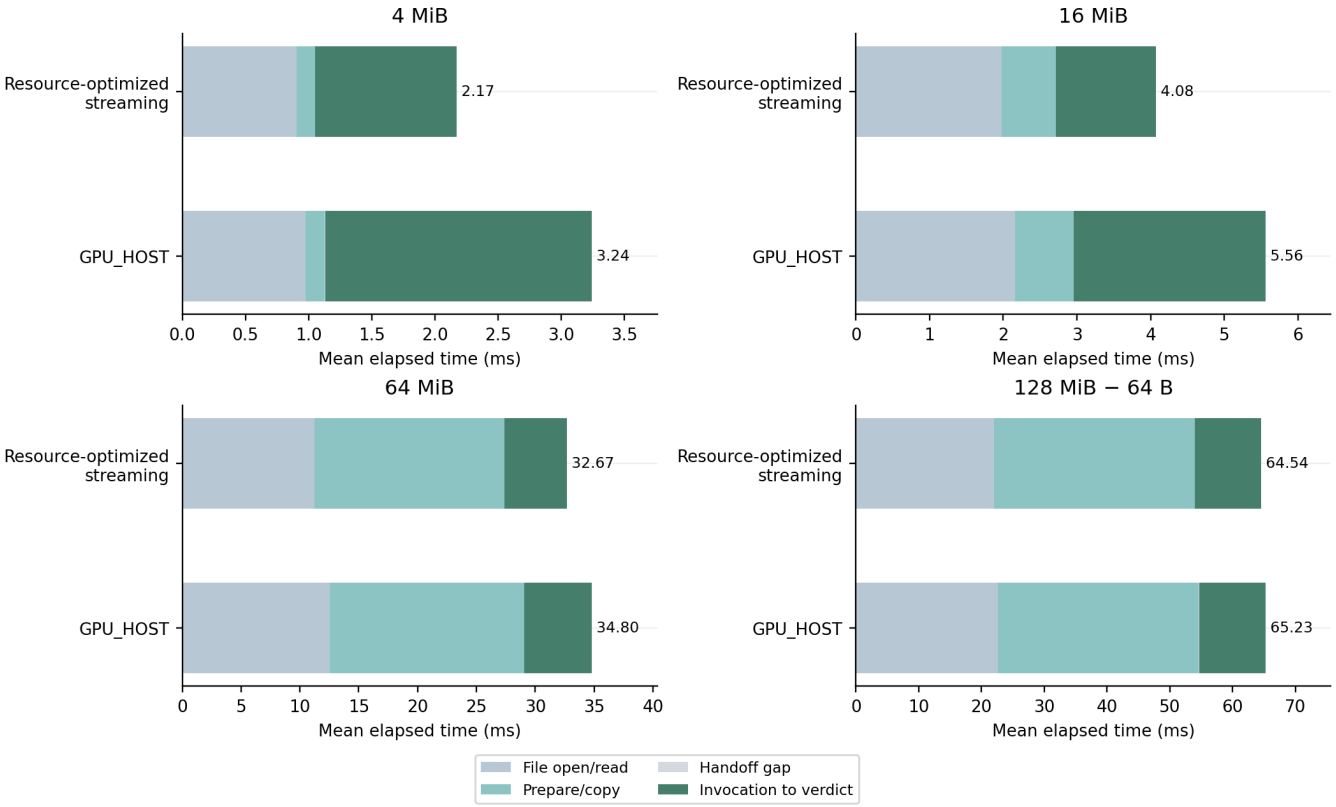
At 4 MiB, Resource-optimized streaming achieves 1.84× host in-memory speedup and 1.50× file-inclusive speedup versus GPU_HOST. At the largest frame, the speedups are 1.00× and 1.01×, respectively.

At the slow boundary, the GPU_HOST / FPGA latency ratio is 0.128. Adjacent sizes must be interpreted separately; do not interpolate across the FPGA preprocessing discontinuity.

The GPU_HOST/FPGA latency ratio uses matching input bytes and timing scope. A value above 1 favors Resource-optimized streaming. Solid synthetic curves and real-input crosses show how the advantage changes with workload size; p95 latency remains visible in the upper panels.

The comparison includes the implemented TensorFlow runtime, transfers, and completion path. It does not predict an ideal fused GPU implementation, and it does not establish equal classification accuracy for FP32 and W8A8.

Where the FPGA/GPU_HOST difference is spent

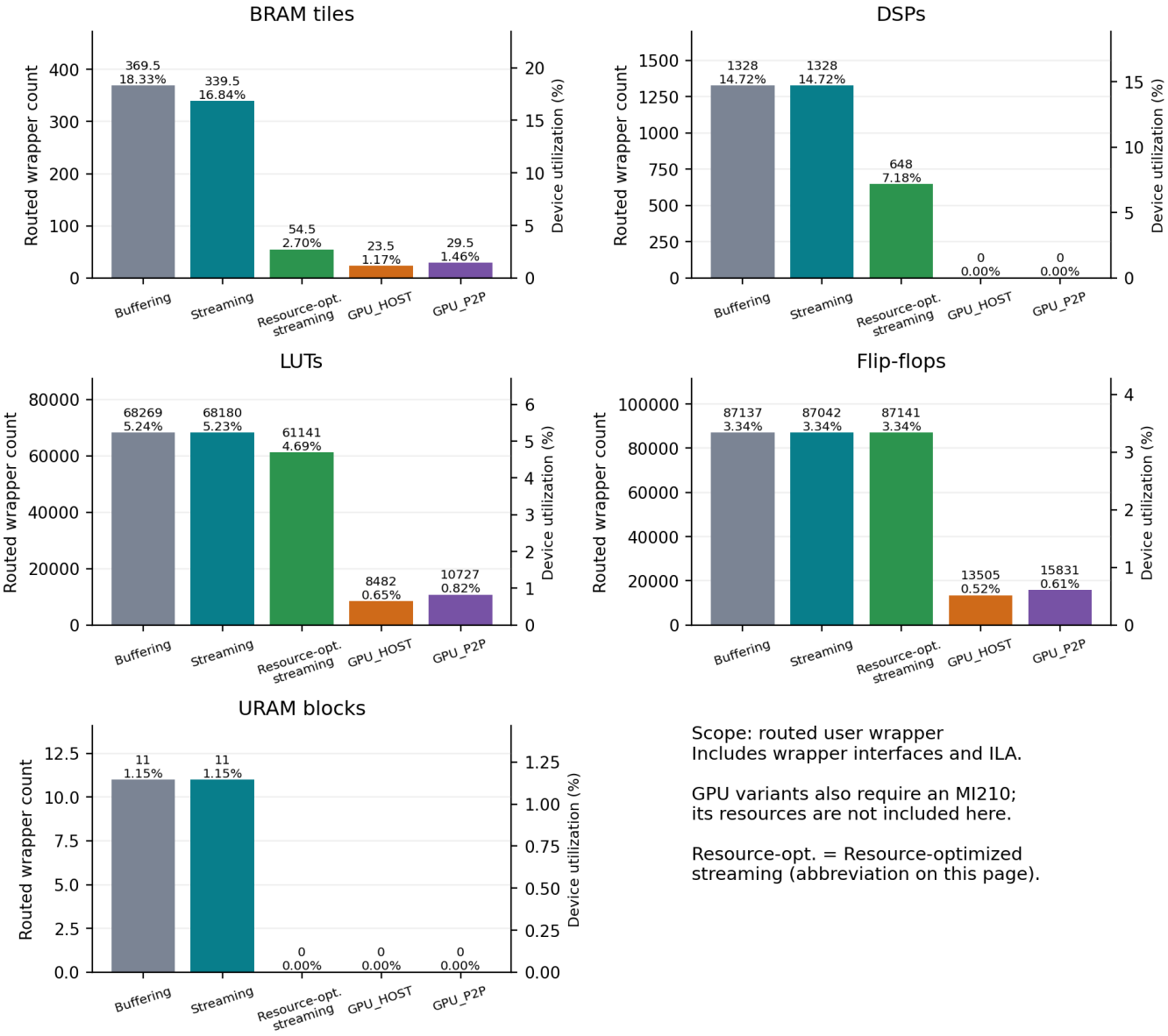


The same additive file-time boundaries are used for both systems: file open/read, host preparation/copy, handoff gap, and invocation to host-observed verdict. Means are plotted so all segments sum exactly. Initial allocation and diagnostic readback remain outside the measured endpoint.

At large sizes, file read and preparation costs dilute the system-level effect of faster inference. At small sizes, the invocation segment contributes more strongly to the FPGA advantage. Apparent differences in file segments are measured host effects; they are not attributable to the FPGA optimization itself.

For the GPU invocation subphases, refer to page 4. For FPGA overlap inside invocation, refer to pages 9–10. Combining their overlapping internal intervals with this outer stack would double-count time.

Resource savings and device utilization



Scope: routed user wrapper
Includes wrapper interfaces and ILA.

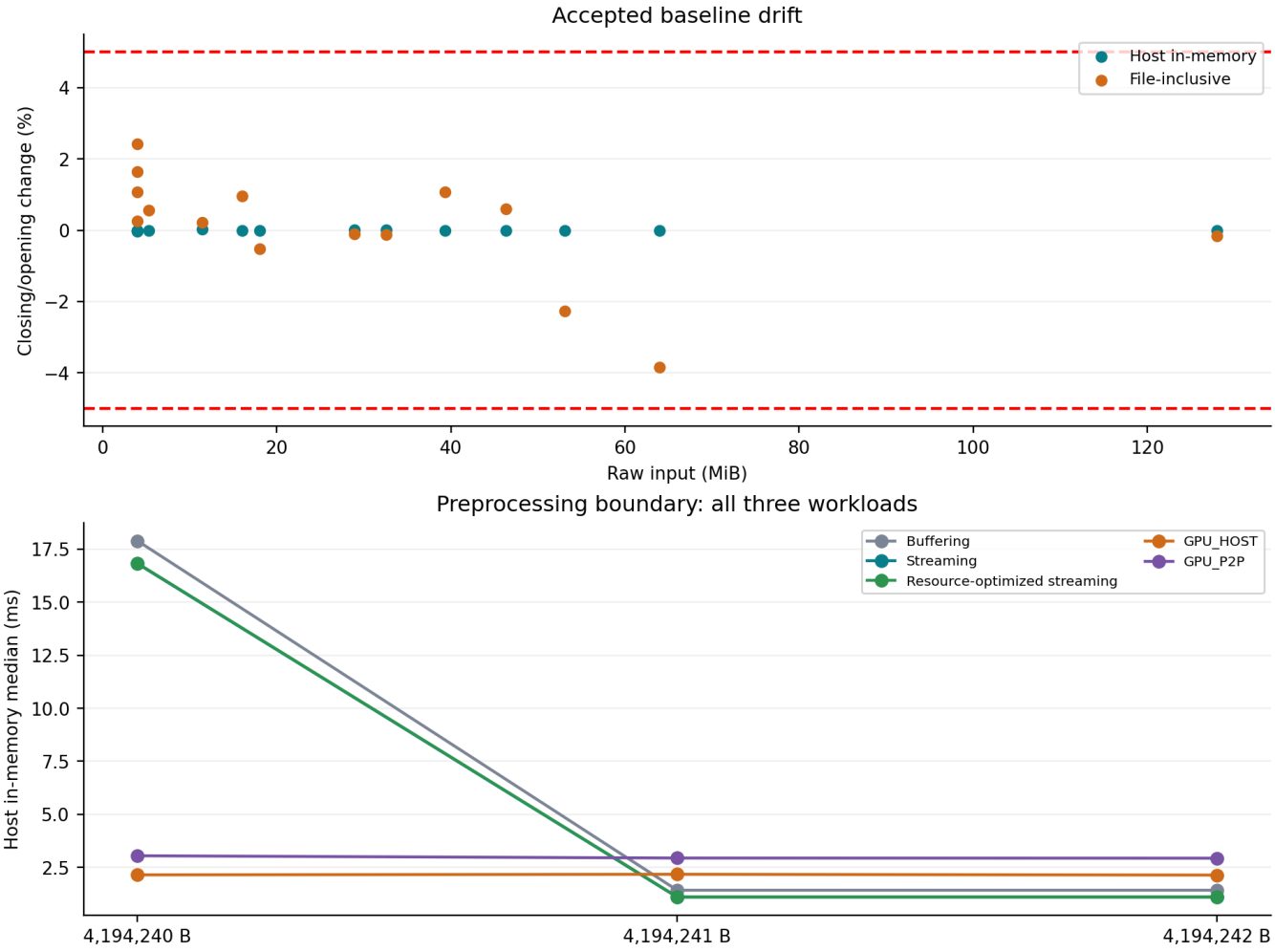
GPU variants also require an MI210;
its resources are not included here.

Resource-opt. = Resource-optimized streaming (abbreviation on this page).

Resource-optimized streaming changes wrapper BRAM from 339.5 to 54.5 tiles (–83.95%), DSPs from 1328 to 648 (–51.20%), and URAM from 11 to 0. BRAM device utilization falls from 16.84% to 2.70%; DSP utilization falls from 14.72% to 7.18%.

Bars and right axes describe the routed user wrapper, including interfaces and ILA. Full-design counts include shell overhead and appear on page 18. GPU-side FPGA resource savings do not account for the MI210's compute, memory, power, or cost.

Validation, drift, and boundary workloads



Maximum accepted closing-baseline drift is 3.84%. Any flagged workloads are remeasured across all variants at the original allocation capacity; evidence and cohorts identify every accepted comparison. Rejected measurements are preserved.

The boundary panel shows a repeatable latency discontinuity, not a correctness failure: at 4,194,240 bytes, Streaming and Resource-optimized streaming spend about 16.777 ms in preprocessing; at 4,194,241 bytes they return to approximately 1.1 ms host in-memory E2E. GPU paths do not show the same discontinuity. Isolating the implementation cause requires separate investigation.

Accepted rows explicitly identify their source attempt and cohort. No warmup, rejected attempt, or diagnostic profiling call enters headline latency distributions. The new campaign does not reuse old latency samples.

FPGA board scores match quantized references exactly. GPU scores match CPU FP32 references within 1e-4 absolute/relative tolerance, with exact verdicts. The 520-call per-image admission, timestamp gates, and file hashes passed; these are execution checks, not a new held-out classification accuracy study.

Complete host in-memory latency table

Each cell: median / p95 in milliseconds; 200 measured trials after 20 warmups. B = boundary, R = real, S = synthetic. Labels are abbreviated only in this dense table: Resource-opt. streaming = Resource-optimized streaming.

Input bytes / type	Buffering	Streaming	Resource-opt. streaming	GPU_HOST	GPU_P2P
4,194,240 / B	17.880 / 17.882	16.807 / 16.808	16.824 / 16.825	2.155 / 2.314	3.053 / 3.326
4,194,241 / B	1.431 / 1.433	1.104 / 1.105	1.119 / 1.121	2.181 / 2.460	2.946 / 3.176
4,194,242 / B	1.431 / 1.436	1.104 / 1.105	1.119 / 1.121	2.145 / 2.260	2.939 / 3.180
4,194,304 / S	1.431 / 1.436	1.104 / 1.105	1.119 / 1.121	2.054 / 2.153	2.894 / 3.121
5,540,996 / R	1.536 / 1.537	1.104 / 1.105	1.121 / 1.121	2.151 / 2.249	3.218 / 3.475
12,004,440 / R	2.041 / 2.042	1.104 / 1.105	1.119 / 1.120	2.728 / 2.899	4.904 / 5.131
16,777,216 / S	2.414 / 2.416	1.342 / 1.344	1.358 / 1.359	2.757 / 2.959	6.137 / 6.359
18,876,484 / R	2.578 / 2.579	1.506 / 1.509	1.522 / 1.523	2.774 / 3.027	6.620 / 6.852
30,323,588 / R	3.472 / 3.473	2.400 / 2.400	2.415 / 2.417	3.087 / 3.211	9.513 / 9.777
34,162,708 / R	3.772 / 3.773	2.699 / 2.700	2.716 / 2.716	3.288 / 3.571	10.604 / 10.847
41,278,872 / R	4.328 / 4.329	3.255 / 3.256	3.271 / 3.272	3.598 / 3.927	12.308 / 12.555
48,540,364 / R	4.895 / 4.897	3.823 / 3.823	3.838 / 3.840	3.942 / 4.122	14.221 / 14.448
55,676,772 / R	5.453 / 5.454	4.380 / 4.381	4.395 / 4.397	4.399 / 4.481	16.050 / 16.264
67,108,864 / S	6.346 / 6.347	5.273 / 5.274	5.290 / 5.290	5.295 / 5.298	18.925 / 19.123
134,217,664 / S	11.589 / 11.590	10.516 / 10.517	10.532 / 10.533	10.537 / 10.543	35.571 / 35.842

Host in-memory input starts at the host invocation. File-inclusive timing includes warm-cache open/read and preparation. Results are selected by case ID, scope, and accepted evidence path, not by taking the newest attempt blindly.

Complete file-inclusive latency table

Each cell: median / p95 in milliseconds; 200 measured trials after 20 warmups. B = boundary, R = real, S = synthetic. Labels are abbreviated only in this dense table: Resource-opt. streaming = Resource-optimized streaming.

Input bytes / type	Buffering	Streaming	Resource-opt. streaming	GPU_HOST	GPU_P2P
4,194,240 / B	19.014 / 19.180	17.918 / 18.080	17.978 / 18.131	3.294 / 3.578	4.118 / 4.428
4,194,241 / B	2.488 / 2.691	2.136 / 2.289	2.166 / 2.333	3.292 / 3.714	4.087 / 4.362
4,194,242 / B	2.485 / 2.790	2.158 / 2.315	2.158 / 2.350	3.230 / 3.560	4.120 / 4.441
4,194,304 / S	2.474 / 2.651	2.154 / 2.287	2.158 / 2.357	3.234 / 3.467	4.075 / 4.307
5,540,996 / R	2.688 / 2.989	2.237 / 2.405	2.252 / 2.413	3.479 / 3.771	4.574 / 4.807
12,004,440 / R	4.030 / 4.814	3.017 / 3.174	3.041 / 3.201	4.632 / 4.960	7.285 / 7.610
16,777,216 / S	5.134 / 5.495	4.010 / 4.319	4.050 / 4.437	5.535 / 5.903	9.149 / 9.462
18,876,484 / R	5.786 / 6.031	4.712 / 5.372	4.633 / 4.903	6.031 / 6.455	10.119 / 10.685
30,323,588 / R	7.628 / 7.814	6.606 / 6.929	6.593 / 6.829	7.694 / 8.483	16.114 / 17.048
34,162,708 / R	16.894 / 17.892	15.825 / 16.180	15.785 / 15.973	17.665 / 18.296	24.165 / 24.810
41,278,872 / R	22.813 / 24.504	21.738 / 23.361	22.010 / 23.602	22.898 / 23.922	31.189 / 32.997
48,540,364 / R	22.112 / 23.506	21.105 / 22.605	21.227 / 21.559	22.861 / 24.088	33.256 / 33.788
55,676,772 / R	30.164 / 32.134	29.316 / 31.595	28.618 / 28.888	24.553 / 26.575	36.642 / 37.029
67,108,864 / S	34.783 / 35.428	33.932 / 34.951	32.620 / 33.009	34.816 / 37.003	49.212 / 49.858
134,217,664 / S	65.308 / 69.577	64.295 / 68.593	64.257 / 65.925	64.863 / 68.333	91.557 / 92.145

Host in-memory input starts at the host invocation. File-inclusive timing includes warm-cache open/read and preparation. Results are selected by case ID, scope, and accepted evidence path, not by taking the newest attempt blindly.

All rows identify their accepted cohort in the exported statistics. This campaign passed the closing-baseline drift gate without requiring a recheck.

GPU profiling detail and timestamp definitions

Profile table: 4 MiB and largest frame, median / p95 (ms), 20 synchronized diagnostic trials per row. Full all-workload profile statistics are exported in data/gpu_profiles.json.

Path / size / phase	Kernel sum	Device span	Host synchronized
GPU_HOST / 4 MiB / preprocessing	0.070 / 0.071	0.922 / 1.022	1.539 / 1.684
GPU_HOST / 4 MiB / inference	0.318 / 0.322	1.687 / 1.736	2.486 / 2.621
GPU_HOST / 128 MiB – 64 B / preprocessing	0.871 / 0.876	1.699 / 1.863	2.458 / 3.028
GPU_HOST / 128 MiB – 64 B / inference	0.317 / 0.320	1.695 / 1.997	2.574 / 3.079
GPU_P2P / 4 MiB / preprocessing	0.070 / 0.072	0.881 / 1.108	1.507 / 1.737
GPU_P2P / 4 MiB / inference	0.322 / 0.323	1.579 / 1.595	2.382 / 2.425
GPU_P2P / 128 MiB – 64 B / preprocessing	0.886 / 0.893	1.685 / 1.837	2.415 / 3.071
GPU_P2P / 128 MiB – 64 B / inference	0.319 / 0.324	1.656 / 1.892	2.550 / 2.958

Quantity	Boundary / meaning
File E2E	File open/read start → host-observed verdict
Host in-memory E2E	Host invocation start → host-observed verdict
Host→GPU copy	HIP copy_begin → copy_end; device event duration, no aligned host start
P2P forwarding	FPGA CSR18 → CSR19; first/last forwarded payload timing
Polling window	CSR38 → CSR41; polling begins → result payload arrival
Ready→payload	CSR40 → CSR41; readiness response → payload arrival
FPGA preprocessing	First raw acceptance (CSR8) → last pixel produced (CSR11)
FPGA CNN interval	First CNN input (CSR12) → score (CSR14); includes stalls
GPU device span	First attributed kernel start → last attributed kernel end
Kernel sum	Sum of attributed kernel durations; excludes gaps
Publication host phase	Before score conversion/DLPack → completion of publication call

FPGA timestamps are calibrated against 32 MMIO brackets before and after each block. GPU collectors use at most 20 trials per calibration block. GPU XPlane events are correlated with Trace markers and step_num in the trace time domain, not Python wall-clock timestamps.

Resources, evidence, and reproduction

Full-design routed utilization, including shell overhead. Each cell shows count / percentage of U55C capacity. GPU variants still require the external MI210.

Variant	BRAM	DSP	LUT	FF	URAM
Buffering	511.5 / 25.37%	1328 / 14.72%	189158 / 14.51%	259187 / 9.94%	11 / 1.15%
Streaming	481.5 / 23.88%	1328 / 14.72%	188993 / 14.50%	258992 / 9.93%	11 / 1.15%
Resource-optimized streaming	196.5 / 9.75%	648 / 7.18%	181886 / 13.95%	259143 / 9.94%	0 / 0.00%
GPU_HOST	165.5 / 8.21%	0 / 0.00%	125538 / 9.63%	176994 / 6.79%	0 / 0.00%
GPU_P2P	171.5 / 8.51%	0 / 0.00%	131566 / 10.09%	187929 / 7.21%	0 / 0.00%

Campaign: hls4ml/optimize/runs/final_benchmarks_gpu_preprocess_v2_20261021_030038. Primary sources: summary.json, measurement_manifest.json, campaign.json, environment_manifest.json, and per-attempt observations/derived/calibration files. Bitstreams and runtimes remain pinned to the campaign identities.

Verification: all 129 manifest hashes checked; 180 accepted case/scope/iteration cells with 200 measured calls each. Every stored median/p95 is recomputed. Additive intervals match elapsed time. Timestamp and parity gates were enforced during collection.

Report exports: figures/ contains SVG, PNG, and PDF plots. data/statistics.csv contains means, medians, p95, cohort, and evidence pointers. data/phase_means.csv contains additive breakdown values; data/timelines.json records selected trials and boundaries. data/gpu_profiles.json preserves all diagnostic profile summaries. evidence_manifest.json records source and generated-file hashes.

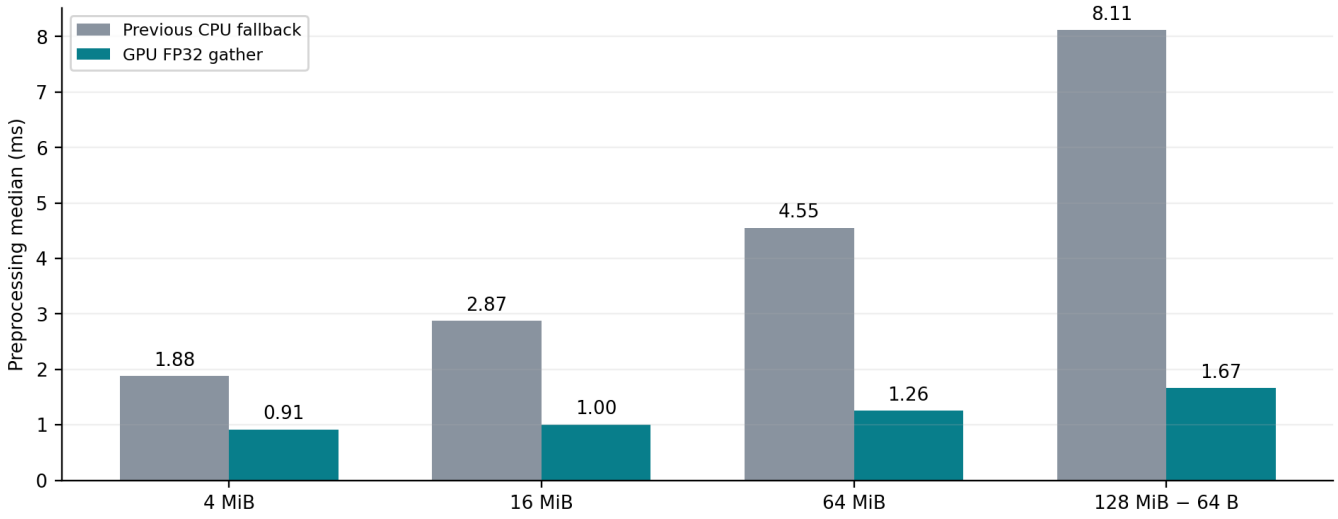
Reproduce: from the repository root, run ml_baseline/.venv/bin/python hls4ml/optimize/reports/optimizations/final_benchmarks_gpu_preprocess_v2_20261021_030038/build_report.py. Uses the existing NumPy/Matplotlib environment and system ReportLab. Reads campaign data and writes only this report directory; no hardware access, rebuild, training, or benchmark replay.

Limits: fixed variant order, one host session, one GPU and FPGA pairing, warm filesystem cache, and different inference precision. File rechecks address observed drift without proving complete stationarity. Resource totals are synthesis/route evidence; latency values are board measurements. No power, energy, cold-storage, concurrent-load, or accuracy-equivalence claims are made.

Resource-optimized streaming retains reduced wrapper BRAM and DSP usage. Latency conclusions use the newly measured preprocessing-v2 runtime only. Subsequent GPU runtime experiments are archived separately and are not selected into this report.

Preprocessing-v2: corrected device execution

The earlier uint8 gather copied the entire raw input from GPU memory to CPU, sampled on CPU, then copied selected bytes back to GPU. This revision casts bytes to FP32 on GPU before gathering. Pixel semantics and model weights are unchanged. The largest input requires an approximately 512 MiB temporary tensor.



Paired software diagnostics: same GPU, inputs, and affinity; 20 warmups and 200 calls per size/revision. Timing includes synchronization and image output materialization. These values are not components to add to unprofiled end-to-end latency.

GPU_BASELINE.pdf reports 0.165 ms preprocessing plus inference, but exact original samples/runtime and timing endpoints are not established. A separate investigation compares explicit scopes and records further runtime revisions. Those later results are deliberately excluded here.